

# Manual CIF preparation for Quantum ESPRESSO

A step-by-step lesson using sodium chloride

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# Lesson goals

By the end of this lesson, the pupil can:

- Identify cell parameters and atom sites in a CIF.
- Generate a full unit cell from symmetry operations.
- Write cell vectors and fractional positions in QE format.
- Check an input file before running `pw.x`.

Worked example: the conventional cubic cell of NaCl.

# The conversion workflow

- ➊ Preserve the source and extract the structural data.
- ➋ Read the cell parameters and the atom table.
- ➌ Expand symmetry, wrap coordinates, and remove duplicates.
- ➍ Write cell vectors and count atoms and species.
- ➎ Add pseudopotentials and calculation settings.
- ➏ Check, save, and run the input.

## Step 1: working files

- Keep the original `NaCl.cif` unchanged.
- Create a plain-text file called `NaCl.in`.
- Copy the structural information into your working notes.
- Keep the source citation with your teaching materials.

For this basic input, leave publication metadata, refinement statistics, and displacement parameters in the source CIF.

## Step 2: the six cell parameters

```
_cell_length_a      5.62  
_cell_length_b      5.62  
_cell_length_c      5.62  
_cell_angle_alpha   90  
_cell_angle_beta     90  
_cell_angle_gamma    90
```

- Lengths:  $a, b, c$ , in Å.
- Angles:  $\alpha = \angle(b, c)$ ,  $\beta = \angle(a, c)$ ,  $\gamma = \angle(a, b)$ , in degrees.
- Here, equal lengths and right angles give cubic cell geometry.

## Step 2: numerical cleanup

| CIF value | Value to use | Meaning                       |
|-----------|--------------|-------------------------------|
| 5.620(3)  | 5.620        | Parentheses give uncertainty  |
| 0.2500(2) | 0.2500       | Keep the central value        |
| ?         | Investigate  | Unknown value                 |
| .         | Investigate  | Inapplicable or missing value |

Required cell lengths and coordinates need valid numbers.

Never substitute zero for an unknown value.

## Step 3: reading a CIF loop

The tags after loop\_ define the column order.

```
loop_  
_atom_site_label  
_atom_site_type_symbol  
_atom_site_symmetry_multiplicity  
_atom_site_Wyckoff_symbol  
_atom_site_fract_x  
_atom_site_fract_y  
_atom_site_fract_z  
_atom_site_occupancy  
_atom_site_attached_hydrogens  
_atom_site_calc_flag  
Na1 Na1+ 4 a 0.0 0.0 0.0 1.0 0 d  
Cl1 Cl1- 4 b 0.5 0.5 0.5 1.0 0 d
```

### Step 3: the structural information

| Site | Element | x   | y   | z   | Occupancy |
|------|---------|-----|-----|-----|-----------|
| Na1  | Na      | 0   | 0   | 0   | 1         |
| Cl1  | Cl      | 0.5 | 0.5 | 0.5 | 1         |

- Use Na and Cl as the QE species labels.
- The CIF labels Na1+ and Cl1- describe oxidation states.
- Multiplicity 4 means four equivalent atoms per listed site here.

# Fractional coordinates

A fractional position  $(x, y, z)$  describes

$$\mathbf{r} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c}.$$

For this cubic cell:

$(0, \frac{1}{2}, \frac{1}{2})$  corresponds to  $(0, 2.81, 2.81)$  Å.

- Fractional coordinates are dimensionless.
- Their meaning depends on the chosen cell vectors.
- Keep fractional coordinates for `ATOMIC_POSITIONS crystal`.

Source: Quantum ESPRESSO `pw.x` input reference

## Step 4: symmetry operations

Find one of these CIF tags:

```
_symmetry_equiv_pos_as_xyz  
_space_group_symop_operation_xyz
```

Examples from the NaCl CIF include:

```
x,y,z  
x,1/2+y,1/2+z  
1/2+x,y,1/2+z  
1/2+x,1/2+y,z
```

Apply every listed operation to every listed atom site.

## Step 4: one worked operation

Operation:

$$(x, y, z) \mapsto (x, \frac{1}{2} + y, \frac{1}{2} + z)$$

For sodium:

$$(0, 0, 0) \mapsto (0, \frac{1}{2}, \frac{1}{2})$$

For chlorine:

$$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \mapsto (\frac{1}{2}, 1, 1)$$

The chlorine result lies on periodic boundaries. Its wrapped position is  $(\frac{1}{2}, 0, 0)$ .

## Step 5: wrapping into the unit cell

For each fractional component  $u$ , use

$$u_{\text{wrapped}} = u - \lfloor u \rfloor, \quad 0 \leq u_{\text{wrapped}} < 1.$$

| Before wrapping | After wrapping |
|-----------------|----------------|
| 1.00            | 0.00           |
| 1.25            | 0.25           |
| -0.25           | 0.75           |
| -1.10           | 0.90           |

## Step 6: duplicate removal

- Many symmetry operations give the same position for a special site.
- Wrap positions first, then retain each distinct atom once.
- For rounded data, compare coordinates with a small tolerance.
- Compare distances across periodic boundaries as well.

Different species at the same position require investigation.  
They may represent a mixed-occupancy site.

## Step 6: the complete NaCl cell

```
Na 0.0 0.0 0.0
Na 0.0 0.5 0.5
Na 0.5 0.0 0.5
Na 0.5 0.5 0.0
```

```
Cl 0.5 0.5 0.5
Cl 0.5 0.0 0.0
Cl 0.0 0.5 0.0
Cl 0.0 0.0 0.5
```

$$4 \text{ Na} + 4 \text{ Cl} = 8 \text{ atoms}$$

The conventional cell contains four NaCl formula units, matching  $Z = 4$ .

# Occupancy and disorder

- Occupancy 1 means a fully occupied crystallographic site.
- Occupancy 0.5 describes an average occupation of half that site.
- An ordinary explicit QE atom list places a whole atom on each row.
- Choose an ordered configuration or an appropriate supercell to model disorder or vacancies.

Our NaCl example has occupancy 1 for both sites.

## Step 7: explicit cell vectors

Use `ibrav = 0` and supply three vectors, one per row:

```
CELL_PARAMETERS angstrom
5.62  0.00  0.00
0.00  5.62  0.00
0.00  0.00  5.62
```

- This diagonal form applies to our orthogonal cubic cell.
- `angstrom` states the units explicitly.
- Keep the eight conventional-cell atom positions with these vectors.

Source: Quantum ESPRESSO `pw.x` input reference

## Step 7: a cell with non-right angles

For a general cell, choose the following Cartesian orientation:

$$\mathbf{a} = (a, 0, 0),$$

$$\mathbf{b} = (b \cos \gamma, b \sin \gamma, 0),$$

$$c_x = c \cos \beta,$$

$$c_y = c \frac{\cos \alpha - \cos \beta \cos \gamma}{\sin \gamma},$$

$$c_z = \sqrt{c^2 - c_x^2 - c_y^2}, \quad \mathbf{c} = (c_x, c_y, c_z).$$

Convert degrees to radians when your calculator or code requires radians.

## Step 8: atom counts and species

```
! In &SYSTEM
nat = 8
ntyp = 2

ATOMIC_SPECIES
Na  22.98977  Na.UPF
Cl  35.45     Cl.UPF
```

- nat: number of explicit atom rows.
- ntyp: number of species labels.
- Each species row gives label, mass, and pseudopotential filename.

Source: Quantum ESPRESSO pw.x input reference

## Step 9: calculation choices

| Choice   | Purpose                                   |
|----------|-------------------------------------------|
| scf      | Solve electrons at a fixed structure      |
| relax    | Optimize atomic positions at a fixed cell |
| vc-relax | Optimize positions and the cell           |

- Start this lesson with `scf`.
- Choose compatible pseudopotentials and appropriate energy cutoffs.
- Choose k-point sampling and test numerical convergence.

Source: Quantum ESPRESSO `pw.x` input reference

## Step 9: k-point sampling

```
K_POINTS automatic  
4 4 4 0 0 0
```

- The first three integers set the mesh along reciprocal directions.
- The last three control half-grid shifts, each either 0 or 1.
- Here all shifts are zero.
- Compare denser meshes until the quantity of interest stabilizes.

This  $4 \times 4 \times 4$  mesh is a teaching starting point. Source: Quantum ESPRESSO pw.x input reference

## Step 10: the input header

```
&CONTROL
  calculation = 'scf'
  prefix = 'NaCl'
  pseudo_dir = './pseudo'
  outdir = './tmp'
/
&SYSTEM
  ibrav = 0
  nat = 8
  ntyp = 2
  ecutwfc = 50
  ecutrho = 400
  occupations = 'fixed'
/
```

```
&ELECTRONS
  conv_thr = 1.0d-8
/
```

Cutoffs are in Ry. These values are illustrative.

fixed occupations suit this insulating example.

Each namelist ends with `/`.

## Step 10: the structure cards

```
ATOMIC_SPECIES
Na 22.98977 Na.UPF
Cl 35.45      Cl.UPF
```

```
ATOMIC_POSITIONS crystal
Na 0.0 0.0 0.0
Na 0.0 0.5 0.5
Na 0.5 0.0 0.5
Na 0.5 0.5 0.0
Cl 0.5 0.5 0.5
Cl 0.5 0.0 0.0
Cl 0.0 0.5 0.0
Cl 0.0 0.0 0.5
```

```
K_POINTS automatic
4 4 4 0 0 0
```

```
CELL_PARAMETERS angstrom
5.62 0.00 0.00
0.00 5.62 0.00
0.00 0.00 5.62
```

Append these cards after the namelists.

Replace both UPF filenames before running.

## Step 11: pre-run checks

- ➊ Eight position rows match `nat` = 8.
- ➋ Two species rows match `ntyp` = 2.
- ➌ Four Na and four Cl atoms match the source composition.
- ➍ No duplicate atoms or unexplained overlaps remain.
- ➎ Cell vectors and fractional coordinates use the same basis.
- ➏ Both UPF files exist in `./pseudo`.

For this ideal structure, the nearest Na–Cl distance is  $a/2 = 2.81 \text{ \AA}$ .

## Step 12: the first run

Run from the folder containing `NaCl.in` and `pseudo/`:

```
mkdir -p tmp  
pw.x -in NaCl.in > NaCl.out
```

Inspect `NaCl.out` for:

- The expected cell and atom counts.
- Successful electronic convergence.
- A final energy and successful completion.

A completed calculation still needs cutoff and k-point convergence tests.

# Pupil exercise

Use the operation  $(x, y, z) \mapsto (x, \frac{1}{2} + y, \frac{1}{2} + z)$ .

- 1 Apply it to Cl at  $(0.5, 0.5, 0.5)$ , then wrap the result.
- 2 Wrap the point  $(-0.25, 1.25, 1.00)$ .
- 3 Explain why the NaCl input uses `nat = 8`.
- 4 Find the error: eight position rows with `nat = 2`.
- 5 State what must change when replacing the conventional cell with a primitive cell.

## Exercise answers

- ①  $(0.5, 1, 1)$ , which wraps to  $(0.5, 0, 0)$ .
- ②  $(0.75, 0.25, 0)$ .
- ③ Symmetry produces four Na atoms and four Cl atoms.
- ④ For these eight explicit rows, `nat` must equal 8.
- ⑤ Transform the cell vectors and atom coordinates together, remove periodic equivalents, and update the atom count and sampling.

# References and teaching files

- Quantum ESPRESSO: `pw.x` input reference.
- CIF conventions: IUCr crystallographic information framework.
- NaCl source: Crystallography Open Database entry 1000041.
- Abrahams and Bernstein (1965), *Acta Crystallographica* 18, 926–932. DOI: 10.1107/S0365110X65002244.

Files: `cif-to-qe-beamer.tex`, `NaCl.cif`, `NaCl.in`.

The LaTeX source includes teaching notes for each slide.